INCH-POUND
MIL-DTL-24643/26G
27 November 2023
SUPERSEDING
MIL-DTL-24643/26F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +302 DEGREES FAHRENHEIT (+150 DEGREES CELSIUS), 600 VOLTS, TYPES LSDPS, LSTPS, LSFPS, LS7PS, LSDPSN, LSTPSN, LSFPSN, AND LS7PSN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in the following variations: LSDPS, LSTPS, LSFPS, and LS7PS (watertight with circuit integrity), and LSDPSN, LSTPSN, LSFPSN, and LS7PSN (non-watertight with circuit integrity).

Construction
Types LSDPS, LSTPS, LSFPS, and LS7PS (Watertight with Circuit Integrity)
Types LSDPSN, LSTPSN, LSFPSN, and LS7PSN (Non-Watertight with Circuit Integrity)

First	-	Copper conductor, nickel-coated (see table I or table II for size and stranding).
Second	-	Extruded silicone rubber insulation (see table I or table II for dimensions).
Third	-	Glass braid (see table I or table II for dimensions). Standard identification code applied by method 4.
Fourth	-	Transparent braid covering.
Fifth	-	The specified number of singles (see table I or table II) shall be cabled together with a lay not greater than 10 inches. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Sixth	-	A binder applied helically with overlap.
Seventh	-	Silicone rubber jacket colored orange or red (see table I or table II for thickness). Jacket surface markings not applicable.
Eighth	-	Braided metal armor.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details (watertight).

Military part no. M24643/26	Type and size	Conductors				Overall cable (inch)		Conductor resistance (ohms)
		No.	AWG (size and stranding)	Diameter over insulation nom. (inch)	Diameter over braid covering max. (inch)	Jacket thickness (min. avg.)	Diameter (max.)	
-01AN	LSDPS-3	2	16 (class B)	0.100	0.125	0.038	0.455	5.39
-02AN	LSDPS-4	2	14 (class B)	0.116	0.141	0.038	0.489	3.38
-03AN	LSDPS-6	2	12 (class B)	0.143	0.170	0.050	0.585	2.13
-04AN	LSDPS-9	2	10 (class B)	0.166	0.193	0.050	0.628	1.34
-05AN	LSDPS-14	2	9 (class B)	0.200	0.230	0.063	0.730	1.06
-06AN	LSTPS-3	3	16 (class B)	0.100	0.125	0.038	0.475	5.39
-07AN	LSTPS-4	3	14 (class B)	0.116	0.141	0.050	0.553	3.38
-08AN	LSTPS-6	3	12 (class B)	0.143	0.170	0.050	0.620	2.13
-09AN	LSTPS-9	3	10 (class B)	0.166	0.193	0.050	0.657	1.34
-10AN	LSTPS-14	3	9 (class B)	0.200	0.230	0.063	0.751	1.06
-11AN	LSTPS-23	3	7 (class B)	0.257	0.291	0.063	0.866	0.643
-12AN	LSTPS-30	3	5 (class C)	0.310	0.348	0.075	0.989	0.403
-13AN	LSFPS14	4	9 (class B)	0.200	0.230	0.063	0.815	1.06
-14AN	LS7PS-6	7	12 (class B)	0.143	0.170	0.063	0.775	2.13
-15AN	LS7PS-14	7	9 (class B)	0.200	0.230	0.063	0.986	1.06

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TABLE II. Details (non-watertight).

Military part no. M24643/26	Type and size	Conductors				Overall cable (inch)		Conductor resistance (ohms)
		No.	AWG (size and stranding)	Diameter over insulation nom. (inch)	Diameter over braid covering max.	Jacket thickness (min. avg.)	Diameter (max.)	
-16AN	LSDPSN-3	2	16 (class B)	0.100	0.125	0.038	0.455	5.39
-17AN	LSDPSN-4	2	14 (class B)	0.116	0.141	0.038	0.489	3.38
-18AN	LSDPSN-6	2	12 (class B)	0.143	0.170	0.050	0.585	2.13
-19AN	LSDPSN-9	2	10 (class B)	0.166	0.193	0.050	0.628	1.34
-20AN	LSDPSN-14	2	9 (class B)	0.200	0.230	0.063	0.730	1.06
-21AN	LSTPSN-3	3	16 (class B)	0.100	0.125	0.038	0.475	5.39
-22AN	LSTPSN-4	3	14 (class B)	0.116	0.141	0.050	0.553	3.38
-23AN	LSTPSN-6	3	12 (class B)	0.143	0.170	0.050	0.620	2.13
-24AN	LSTPSN-9	3	10 (class B)	0.166	0.193	0.050	0.657	1.34
-25AN	LSTPSN-14	3	9 (class B)	0.200	0.230	0.063	0.751	1.06
-26AN	LSTPSN-23	3	7 (class B)	0.257	0.291	0.063	0.866	0.643
-27AN	LSTPSN-30	3	5 (class C)	0.310	0.348	0.075	0.989	0.403
-28AN	LSFPSN-14	4	9 (class B)	0.200	0.230	0.063	0.815	1.06
-29AN	LS7PSN-6	7	12 (class B)	0.143	0.170	0.063	0.775	2.13
-30AN	LS7PSN-14	7	9 (class B)	0.200	0.230	0.063	0.986	1.06

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I or table II
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

NOTE: The nickel coating on the conductors of the completed cable shall be permitted to have minor nicks, scrapes, or abrasions that are just perceptible to the eye.

Watertightness (types LSDPS, LSTPS, LSFPS, and LS7PS only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (insulation, scrapes min.)	250
Crack resistance	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (watertight types only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	800
Elongation (percent, min.)	250
Durometer (jacket) – type A (hardness, min.)	60

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Group C:

Physicals (cable aging) (500±18 °F [260±10 °C])	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	600
Elongation (percent, min.)	150
Cable fill sealant removability (watertight types only)	No failure
Armor (conformance to material, construction, and coverage)	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Armor	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Smoke index (max.)	
Jacket	35
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5

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Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (min.)
All types and sizes	100

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-042)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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